

## SAFETY DATA SHEET

Creation Date 26-September-2009

Revision Date 29-March-2024

Revision Number 6

### 1. Identification

**Product Name** Nitromethane d-(3)

**Cat No. :** 42338

**CAS-No** 13031-32-8  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                  |            |
|--------------------------------------------------|------------|
| <b>Flammable liquids</b>                         | Category 3 |
| <b>Acute oral toxicity</b>                       | Category 4 |
| <b>Acute Inhalation Toxicity</b>                 | Category 4 |
| <b>Carcinogenicity</b>                           | Category 2 |
| <b>Reproductive Toxicity</b>                     | Category 2 |
| <b>Physical Hazards Not Otherwise Classified</b> | Category 1 |
| Risk of explosion if heated under confinement    |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable liquid and vapor  
Harmful if swallowed or if inhaled  
Suspected of causing cancer  
Suspected of damaging fertility or the unborn child  
Risk of explosion if heated under confinement

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Wear protective gloves/protective clothing/eye protection/face protection  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep cool. Protect from sunlight  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Use non-sparking tools  
Take action to prevent static discharges

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Rinse mouth  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
IF exposed or concerned: Get medical advice/attention  
Explosion risk in case of fire  
IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

**Storage**

Store locked up  
Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component         | CAS-No     | Weight % |
|-------------------|------------|----------|
| Nitro(2H3)methane | 13031-32-8 | 100      |
| Nitromethane      | 75-52-5    | -        |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                        |                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                         |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                              |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                                              |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: May cause methemoglobinemia |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                     |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 35 °C / 95 °F                                                                                                                           |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                                                       |
| <b>Lower</b>                            | No data available                                                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
2

**Instability**  
3

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

precautionary measures against static discharges.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep under nitrogen. Protect from moisture. Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Acids. Bases. Strong acids. Amines. Aldehydes. Ketones. Organic acids. Lead. Acetone. Metals. copper. Reducing Agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component    | Alberta                                  | British Columbia | Ontario TWAEV | Quebec                                     | ACGIH TLV   | OSHA PEL                                                                                                     | NIOSH         |
|--------------|------------------------------------------|------------------|---------------|--------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------|---------------|
| Nitromethane | TWA: 20 ppm<br>TWA: 50 mg/m <sup>3</sup> | TWA: 20 ppm      | TWA: 20 ppm   | TWA: 100 ppm<br>TWA: 250 mg/m <sup>3</sup> | TWA: 20 ppm | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 250 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 250 mg/m <sup>3</sup> | IDLH: 750 ppm |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Protective gloves

| Glove material              | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before

re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                          |
|----------------------------------------|------------------------------------------|
| Physical State                         | Liquid                                   |
| Appearance                             | Colorless                                |
| Odor                                   | No information available                 |
| Odor Threshold                         | No information available                 |
| pH                                     | No information available                 |
| Melting Point/Range                    | No data available                        |
| Boiling Point/Range                    | 100 - 102 °C / 212 - 215.6 °F @ 760 mmHg |
| Flash Point                            | 35 °C / 95 °F                            |
| Evaporation Rate                       | No information available                 |
| Flammability (solid,gas)               | Not applicable                           |
| Flammability or explosive limits       |                                          |
| Upper                                  | No data available                        |
| Lower                                  | No data available                        |
| Vapor Pressure                         | No information available                 |
| Vapor Density                          | No information available                 |
| Specific Gravity                       | 1.180                                    |
| Solubility                             | No information available                 |
| Partition coefficient; n-octanol/water | No data available                        |
| Autoignition Temperature               | No information available                 |
| Decomposition Temperature              | No information available                 |
| Viscosity                              | No information available                 |
| Molecular Formula                      | C D3 N O2                                |
| Molecular Weight                       | 64.06                                    |

## 10. Stability and reactivity

|                                  |                                                                                                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                                              |
| Stability                        | Hygroscopic. Stable under normal conditions. Risk of explosion by shock, friction, fire or other sources of ignition.                                                            |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Do not subject to grinding/shock/friction. Excess heat. Incompatible products. Exposure to moist air or water. |
| Incompatible Materials           | Acids, Bases, Strong acids, Amines, Aldehydes, Ketones, Organic acids, Lead, Acetone, Metals, copper, Reducing Agent                                                             |
| Hazardous Decomposition Products | Nitrogen oxides (NO <sub>x</sub> ), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                                                      |
| Hazardous Polymerization         | No information available.                                                                                                                                                        |
| Hazardous Reactions              | None under normal processing.                                                                                                                                                    |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component    | LD50 Oral         | LD50 Dermal            | LC50 Inhalation               |
|--------------|-------------------|------------------------|-------------------------------|
| Nitromethane | 940 mg/kg ( Rat ) | >2000 mg/kg ( Rabbit ) | LC50 = 11.02 mg/L ( Rat ) 1 h |

Toxicologically Synergistic Products No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation May cause skin, eye, and respiratory tract irritation

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component         | CAS-No     | IARC       | NTP                    | ACGIH      | OSHA       | Mexico     |
|-------------------|------------|------------|------------------------|------------|------------|------------|
| Nitro(2H3)methane | 13031-32-8 | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Nitromethane      | 75-52-5    | Group 2B   | Reasonably Anticipated | A3         | X          | A3         |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*Mexico - Occupational Exposure Limits - Carcinogens*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

*A4 - Not Classifiable as a Human Carcinogen*

*A5 - Not Suspected as a Human Carcinogen*

**Mutagenic Effects** No information available

**Reproductive Effects** Category 2.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:  
May cause methemoglobinemia

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component    | Freshwater Algae                                  | Freshwater Fish                                       | Microtox   | Water Flea |
|--------------|---------------------------------------------------|-------------------------------------------------------|------------|------------|
| Nitromethane | EC50: = 36 mg/L, 72h<br>(Desmodesmus subspicatus) | LC50: < 278 mg/L, 96h static<br>(Pimephales promelas) | Not listed | Not listed |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

| Component | log Pow |
|-----------|---------|
|-----------|---------|

|              |      |
|--------------|------|
| Nitromethane | 0.17 |
|--------------|------|

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1261  
 Proper Shipping Name NITROMETHANE  
 Hazard Class 3  
 Packing Group II

#### TDG

UN-No UN1261  
 Proper Shipping Name NITROMETHANE  
 Hazard Class 3  
 Packing Group II

#### IATA

UN-No UN1261  
 Proper Shipping Name NITROMETHANE  
 Hazard Class 3  
 Packing Group II

#### IMDG/IMO

UN-No UN1261  
 Proper Shipping Name NITROMETHANE  
 Hazard Class 3  
 Packing Group II

### 15. Regulatory information

#### International Inventories

| Component         | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Nitro(2H3)methane | 13031-32-8 | -   | -    | -    | -                                             | 235-892-2 | -      | -   |
| Nitromethane      | 75-52-5    | X   | -    | X    | ACTIVE                                        | 200-876-6 | -      | -   |

| Component         | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Nitro(2H3)methane | 13031-32-8 | -     | -        | -    | -    | X    | -    | X     | -     |
| Nitromethane      | 75-52-5    | X     | KE-26005 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant | Canadian Environmental | Canada's Chemicals Management |
|-----------|-----------------------------|------------------------|-------------------------------|
|-----------|-----------------------------|------------------------|-------------------------------|

|              | Release Inventory (NPRI) | Protection Agency (CEPA)<br>- List of Toxic Substances | Plan (CEPA)                                       |
|--------------|--------------------------|--------------------------------------------------------|---------------------------------------------------|
| Nitromethane | Part 4 Substance         |                                                        | Subject to Monitoring and Surveillance Activities |

**Legend**

NPRI - National Pollutant Release Inventory

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component         | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Nitro(2H3)methane | 13031-32-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Nitromethane      | 75-52-5    | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component         | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Nitro(2H3)methane | 13031-32-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Nitromethane      | 75-52-5    | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
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www.thermofisher.com

**Creation Date**

26-September-2009

**Revision Date**

29-March-2024

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**